

Article

In-Situ Test and Numerical Simulation of Anchoring Performance of Embedded Rock GFRP Anchor

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Abstract: Compared to traditional steel reinforcement, GFRP anchors demonstrate outstanding mechanical performance and corrosion resistance, and so they are an ideal substitute for steel reinforcement in anti-floating projects. Based on finite element software, a 3D axisymmetric calculation model of GFRP anti-floating anchors in medium-weathered granite was established in this paper. Combined with the in-situ ultimate pull-out tests, the bonding anchoring performance and bearing characteristics between the anchor body, anchoring mortar, and rock–soil mass were analyzed. The research findings indicated that the cohesive bonding elements exhibited a high degree of conformity in defining the interface contact relationship of the GFRP anti-floating anchor anchoring system. The axial force of the GFRP anti-floating anchor body is “attenuated” along the depth direction, and there was a critical value of anchoring length; under the same conditions, the reasonable anchoring length should be 3.5–5.0 m. All the anchors in the in-situ tests exhibited interfacial shear slip failure between the anchor body and the anchor mortar, with an average maximum load of 450 kN, which is consistent with the maximum failure load of the simulated anchors. Compared to a load of 50 kN, the maximum stress of the anchor mortar increased by 50% under a load of 450 kN. The displacement variation of the surrounding rock–soil mass showed a decreasing trend from the inside to the outside and from the top to the bottom. The research results provided valuable references for the optimization design of GFRP anti-floating anchors.

Keywords: GFRP anti-floating anchor; in-situ ultimate pull-out test; numerical simulation; axial forces; shear stress; critical anchoring length; anchor solid stress



Citation: Hao, Z.; Liu, H.; Yan, N.; Wang, Z.; Bai, X.; Han, J.; Mi, C.; Jia, S.; Sun, G.; Zhu, L.; et al. In-Situ Test and Numerical Simulation of Anchoring Performance of Embedded Rock GFRP Anchor. *Buildings* **2023**, *13*, 2799. <https://doi.org/10.3390/buildings13112799>

Academic Editors: Pengfei Liu and Maria Teresa De Risi

Received: 5 September 2023

Revised: 21 October 2023

Accepted: 6 November 2023

Published: 8 November 2023



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1. Introduction

Fiber-reinforced polymer (FRP) has been widely researched and applied in the field of civil engineering due to its advantages of light weight, high strength, fatigue resistance, and corrosion resistance [1–3]. The traditional FRP bars include glass FRP (GFRP), carbon FRP (CFRP), aramid FRP (AFRP), and basalt FRP (BFRP). However, the price of CFRP and AFRP is relatively expensive, and the durability of BFRP is relatively low. Therefore, with the characteristics of strong durability and low cost, GFRP has become a research hotspot [4–6]. The emergence of GFRP anchors provides new ideas for the choice of structural type in anti-floating projects [7]. These anchors can effectively reduce the weight of structures, improve durability, and simplify the substitution of old materials. However, owing to its anisotropic, non-homogeneous, and linear elastic properties, there is a different anchoring mechanism for GFRP anchors than for conventional steel bars [8].

The bonding strength between the anchoring interface of GFRP anti-floating anchors is mainly provided by chemical bonding, bearing resistance, and frictional resistance [9].

Many researchers have conducted related studies on the material properties and mechanical characteristics of GFRP through laboratory and in-situ tests. Unlike the shear failure of the anchoring mortar and the rock–soil mass interface of conventional steel reinforcement, there is a significant difference in the load transfer mechanism of GFRP anti-floating anchors, which mainly occurs at the anchoring rod–mortar interface [10]. Relevant studies have shown that the M-C failure criterion could well characterize the anchoring performance of the anchor body and the anchoring mortar interface, showing good applicability [11]. To improve the accuracy of test monitoring results, Kou et al. [12] embedded FBG sensors into a GFRP anchor in order to monitor the changes of stress during the pull-out process. Yan et al. [13] applied an Artificial Neural Network and Genetic Algorithm to test the analysis and proposed a new evaluation method for anchoring performance. Fava et al. [14] designed and developed a new anchoring system to improve the reliability of measuring the tensile strength of a large-diameter GFRP anchor. Different anchoring methods and anchoring lengths could affect the anchoring performance of anti-floating anchors [15]. The mechanical anchoring ultimate bearing capacity was 1.08–1.24 times that of straight-line anchoring, and the ultimate bearing capacity of the anchor increased with the increase of anchoring length, but there existed a critical anchoring length [16], and the increase in anchor load capacity was relatively small after exceeding this critical value [17]. The slip tendency of GFRP pull-out specimens under a sustained load was more pronounced than that of steel specimens, and increasing the anchoring length causes greater changes in bond stress [18]. However, under a cyclic load, the bonding strength of GFRP specimens deteriorated more severely compared to steel specimens [19].

The influence of external environmental conditions on the anchoring performance of GFRP anti-floating anchors was also important for long-term support structures [20]. With the increase of temperature, the embedding effect of the anchor bar on the anchoring mortar would weaken [21], and the stiffness of the GFRP reinforcement itself would also decrease [22]. Therefore, Ozkal et al. [23] established an empirical model for the degradation of bonding strength at the anchoring interface with increasing temperature. Biscaia et al. [11] proposed the evolution laws of adhesive strength and the internal friction angle of the anchoring interface under salt spray, wet–dry, and different temperature cycles over time. D’Antino et al. [24] conducted research on the influence of different erosion environments on the tensile properties of GFRP anchors in order to calibrate the environmental reduction factor, where alkaline solutions could lead to a severe loss of bond strength at the anchorage interface [20]. In addition, the different anchoring materials could also affect the anchoring performance of GFRP anti-floating anchors [25]. For example, geopolymer concrete could increase its tensile strength by 45% [26], fiber-reinforced self-compacting light-weight concrete could eliminate concrete splitting failure [27], and steel fiber-reinforced self-compacting concrete could increase the maximum development stress of the anchor [28]. To enrich the research methods, Pepe et al. [28] defined the anchoring behavior of a GFRP anchor through a numerical simulation and revealed the relevant parameters of the bond stress–slip relationship. Safdar et al. [29] proposed a finite element modeling method for different node structures of GFRP anchors in concrete. Sun et al. [30] introduced cohesive bond elements to define the external anchoring model between a GFRP anchor and concrete. Gu et al. [31] conducted a comparative analysis of simulations and test results, and found that the theoretical and finite element methods could accurately predict the bond performance of GFRP anchors, with a relative error of less than 10%. Yan F et al. [32] validated the accuracy of numerical simulation methods based on experimental measurements and presented a bond damage assessment approach for the GFRP anchor–concrete interface.

In sum, the current research on the material properties and mechanical performance of GFRP anti-floating anchors is relatively comprehensive, but has mostly focused on laboratory and in-situ tests, while finite element analysis still has limitations, such as the rough modeling of GFRP anchors, affecting computational accuracy, and relatively limited research on the mechanical properties of anchoring mortar and the surrounding rock–soil

mass which could not be monitored in experiments. It is urgent to carry out numerical simulation research on the anchoring performance of GFRP anti-floating anchors. In view of this, this study established a 3D axisymmetric model of a GFRP anti-floating anchor in weathered granite layers, combined in-situ tests to monitor results, analyzed the evolution laws of stress and deformation of the anchor along the depth direction, and revealed the stress characteristics of the anchoring mortar and surrounding rock–soil mass, in order to improve the anchoring mechanism of GFRP anchors and provide a valuable reference for the design of GFRP anti-floating anchors.

2. In-Situ Ultimate Pull-Out Test

2.1. Test Scheme

2.1.1. Engineering Background

The main rock type in the experimental field is weathered granite, which is reddish in color and has a coarse-grained structure. The rock mass is blocky in structure, with the main mineral components being feldspar and quartz. The total length of the test anchor is located in the weathered granite, with a thickness ranging from approximately 1.6 to 13.7 m and an average layer thickness of 3.45 m. The rock has a specific gravity of 24.5 kN/m³, a uniaxial compressive strength of 32 MPa, an internal friction angle of 55°, a Poisson's ratio of 0.33, a compressibility modulus of 5.0 GPa, and a bearing capacity of 2.5 MPa. The hardness of this rock layer is classified as relatively soft rock, and the rock mass has an embedded fragmented structure, with a degree of integrity classified as relatively fractured. The basic quality grade of the rock mass is IV. According to the geological survey report, the water level in the field is buried at a depth of 2.60~3.80 m.

2.1.2. Implanted FBG

The embedded bare optical fiber grating sensing technology is a process of embedding the bare optical fiber grating in the middle part of the GFRP anchor during processing and shaping, and then pouring it together to form [33]. The bare optical fiber adopts SMF28-C optical fiber with a diameter of about 900 µm. Along the direction of the anchor's design burial depth, the engraving of gratings (equivalent to sensors) were fabricated at positions 0.5, 1.2, 1.9, 2.6, 3.3, 4.0, and 4.7 m away from the anchor hole. A total of seven sensors were installed to monitor the wavelength drift of the grating during the test loading process. Considering the short duration of the applied load in this experiment, temperature changes could be ignored. Therefore, the strain calculation of the anchor in the test could be simplified as Equation (1) [12].

$$\Delta\varepsilon = \frac{\Delta\lambda_B}{K_\varepsilon} \quad (1)$$

where $\Delta\lambda_B$ is the change in the center wavelength of the grating; K_ε is the strain sensitivity coefficient; $\Delta\varepsilon$ is the change in the measured structural strain.

The strain values along the depth direction of the rod obtained from Equation (1) were then converted into axial and shear stresses [10].

2.2. Test Material

2.2.1. GFRP Anti-Floating Anchor

Three GFRP anti-floating anchors with a model YF-H50 were selected for the test. The epoxy resin makes up around 25% of the composition, while the glass fiber makes up the remaining 75%. The anchors have a cross-sectional area of 590 mm², a density of 2.1 g/cm³, and a weight of 1195 g/m. The relevant mechanical parameters of the GFRP anti-floating anchor are shown in Table 1. The anchoring section of the GFRP anchor is fully anchored in the weathered granite layer. The diameter of the test borehole is 110 mm, and the depth of the borehole is 5.5 m, exceeding the internal anchorage length of the anti-floating anchor by 0.5 m. The anchor was anchored to the soil by pouring M30 cement mortar. In order to improve the bond strength and ensure the compact filling of the anchor hole, bottom

grouting and secondary grouting methods were employed. The anchor mortar needed to be cured for 28 days.

Table 1. Mechanical parameters of GFRP anti-floating anchors.

Anchor Model	Anchor Diameter (m)	Total Length (m)	Anchor Length (m)	Ultimate Load (kN)	Tensile Strength (MPa)	Shear Strength (MPa)	Modulus of Elasticity (GPa)
YF-H50	28	6.5	5.0	342	675	150	41

2.2.2. Anchorage

A bonded steel casing was used to protect the GFRP anchor during the testing process. Prior to the start of the experiment, a 1.2 m long steel casing (with an inner diameter of $\varphi = 50$ mm and a wall thickness of $t = 6$ mm) was installed on the exposed section of the anchor at the loading end. The GFRP anchor and the steel casing were tightly bonded together using a mixture of epoxy resin and curing agent. During the tensile testing, the GFRP anchor was securely fixed to the outside of the steel casing using a welded fixture. The current anchorage design ensures that there is no stress concentration phenomenon at the loading end, thus avoiding the material failure of the anchor caused by this phenomenon and its improvement enhances the accuracy of the test results.

2.2.3. Test Equipment

The method of using the Lattice reaction beam for ultimate pull-out testing was implemented as follows: Two buttresses were placed at a distance of 0.5 m from the anchor ends. The lattice reaction beam, piercing jack, anchor dynamometer, and welded anchor were sequentially installed on the buttresses. A 30 mm-thick through steel plate was installed between each component. Subsequently, the welded anchor was welded to the aforementioned steel casing. The research was conducted using a manually operated hydraulic jack, model KQF-60 t, with a stroke of 20 cm. The through steel plate had an area approximately twice that of the jack's cross-section. The load measurement was carried out using an MGH-500 anchor dynamometer and a GSJ-2A type testing instrument manufactured by Shandong University of Science and Technology (Qingdao, China). The strain of the anchor bar was measured using implanted FBG and an SI425 fiber optic grating demodulator. The slip displacement of the anchor bar after each level of loading was measured using a dial gauge with an accuracy of 0.01 mm and a range of 30 mm. The schematic diagram of the test setup is shown in Figure 1.

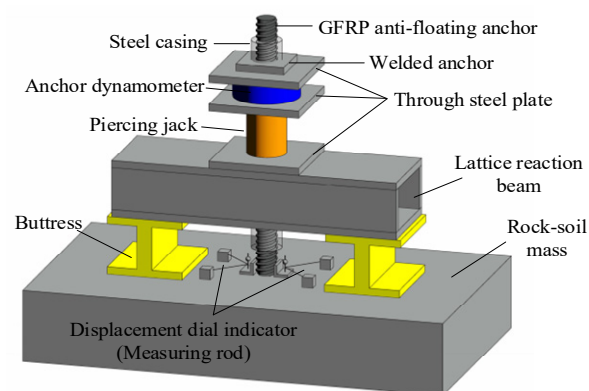


Figure 1. Schematic diagram of anchor test setup.

2.3. Test Procedure

The research was a destructive test, and the loading process was carried out using a staged loading method. The initial load was set at 50 kN, and each subsequent level

increased by 50 kN. The loading sequence was as follows: 0 → 50 kN → 100 kN → 150 kN → 200 kN → 250 kN → 300 kN → ... The loading rate was maintained at 0.2 kN/s until the anchor bar failed. After each level of loading, the displacement was immediately measured, and subsequent measurements were taken every 5 min. The time interval between adjacent levels of loading was 15 min. The anchor was considered to have failed if any of the following conditions occurred (JGJ 476-2019) [34]:

- (1) The displacement increment of the anchor head caused by the current level of loading reaches or exceeds twice the displacement increment caused by the previous level of loading;
- (2) The displacement of the anchor head does not converge;
- (3) Failure of the anchor, with the bar being pulled out from the anchoring body or the anchoring body being pulled out.

The research data in this study were based on previous in-situ ultimate pull-out tests conducted by the research team on GFRP anti-floating anchors [33]. Three GFRP anti-floating anchors produced interfacial shear-slip damage between the anchors and the anchoring mortar. The ultimate load capacities were 450, 400, and 500 kN, respectively. Based on the test results, load-slip curves, axial force distribution curves, and shear stress distribution curves of the anchor were plotted and compared with the results obtained from numerical simulations.

3. Numerical Simulation of Anchorage Performance

3.1. Model Building

3.1.1. Modeling

The numerical simulation in this study was conducted with reference to the in-situ test conditions, using a scaled model of the in-situ test and numerical modeling. Considering the three-dimensional symmetry of the loading and anchoring system models, a 3D axisymmetric model was established for the finite element model of the internal anchoring system using the axisymmetric model in ABAQUS. Only half of the anchoring system was modeled, with the axis of symmetry being the centerline in the longitudinal direction of the anchor. After assigning material properties to the components, the components were assembled. The assembled axisymmetric model of the GFRP anti-floating anchor anchoring system is shown in Figure 2.

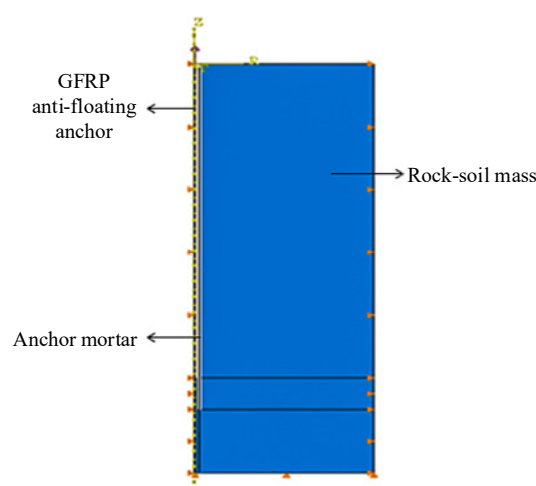


Figure 2. Axisymmetric modeling of GFRP anti-floating anchors.

3.1.2. Material Property Settings

GFRP reinforcement itself is an orthotropic linear elastic material. However, due to the slender nature of the GFRP anti-floating anchor, it was simplified as an isotropic linear elastic material in this simulation. The material type was selected as isotropic, with a

Young's modulus of 41 GPa, a Poisson's ratio of 0.23, and a mass density of 2100 kg/m³. The selection criteria of relevant input parameters were based on the literature reference [30].

The anchoring cement mortar used in this test was selected to utilize the Concrete Damage Plasticity model. It had a strength grade of M30 and a density of approximately 2152 kg/m³. In the simulation, the grouting material was assumed to be an isotropic linear elastic material, with a Young's modulus of 18 GPa and a Poisson's ratio of 0.2.

The elastic modulus of the moderately weathered granite is 3.5 GPa; the specific physico-mechanical parameters are according to Section 2.1.1 "Engineering background". The simulation of the soil-rock mass in this study was conducted using the M-C constitutive model.

The model radius of the GFRP anchor was 0.014 m and the length was 5.0 m; the radius of the anchoring mortar was 0.055 m and the length was 5.5 m; the radius of the surrounding moderately weathered granite body was according to the force of the anchors. Its influence range was not less than 30 D (D is the diameter of the drilling holes of anchors), the model took the range of the surrounding rock-soil mass body 2.0 m, and the depth was 6.5 m.

3.1.3. Interface Property Settings

The interface in this simulation was modeled using cohesive elements and the Maxe damage criterion, which represented the cohesive material between the two parts. The cohesive element could simulate the viscous connection between two parts. When simulating the cohesive interface, the cohesive element assumes the presence of normal stress t_n and shear stresses t_s and t_t on the interface. It could be used to simulate the chemical adhesion, mechanical interlocking, and interface friction between materials, which was similar to the types of forces between the anchor body and anchoring material in the theoretical analysis of the internal anchoring system. Using this cohesive element in the experimental study could simulate interface fracture failure. Initially, during the loading phase, the anchor system undergoes elastic deformation. With the load increased, the cohesive element was damaged, resulting in a decrease in stiffness and interface slip, until the cohesive action was lost, leading to slip failure. This was similar to the actual failure mode observed in in-situ anchor pull-out tests. Therefore, it could be seen that the choice of using cohesive elements provides a high degree of similarity in the simulation.

3.1.4. Boundary Condition Setting

The boundary conditions should be set according to the actual constraints of the anti-floating anchor. Specifically, a vertical constraint should be applied at the bottom surface of the rock-soil mass to prevent vertical displacement. As for the symmetric axis, there was no horizontal displacement at the axis of symmetry. Therefore, a horizontal constraint should be applied to the anchor on one side of the symmetric axis. On the side of the rock-soil mass that was further away from the symmetric axis, horizontal constraints should be applied to account for the influence of the anchor bar's force on not generating horizontal displacement.

3.2. Simulation Process

In the in-situ test simulated in this study, each load level of 50 kN was applied to the loading section of the anchor. During the simulation, each load level of 50 kN was equivalent to a surface load applied at the top of the anchor. Specifically, a surface load of 84.7 MPa was applied at the top of the anchor for each load level. Each load level was set as an analysis step, with the loading sequence as follows: 84.7 MPa → 169 MPa → 254 MPa → 339 MPa → 424 MPa → 508 MPa → 593 MPa → . . . until the anchor failed.

When dividing the mesh, solid homogeneous elements were used, with the anchor body, anchoring mortar, and rock-soil mass elements being of type CAX4R, which were four-node bilinear axisymmetric quadrilateral elements. Reduced integration algorithms and hourglass control methods were used to speed up calculations and avoid units that

had no stiffness and cannot resist deformation. For the cohesive bonding layer elements, when setting the grid control properties, the element shape was set as a quadrilateral, the meshing technique used was sweeping, and the mesh element family was set as cohesive with a value of 0.001. Therefore, the cohesive elements were of type COHAX4, which were four-node axisymmetric cohesive elements. The final mesh division result is shown in Figure 3.

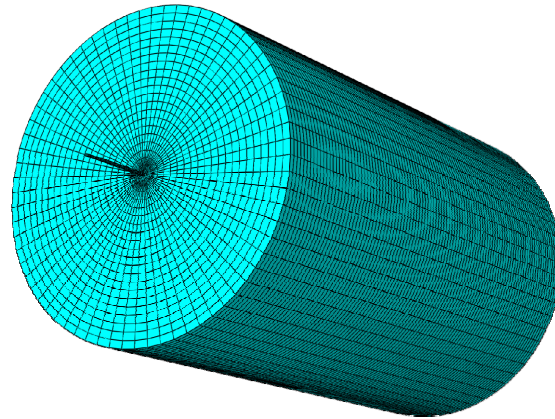


Figure 3. GFRP anti-floating anchor model mesh division.

3.3. Initial Ground Stress Equilibrium Analysis

In order to check whether the initial stress state distribution of the model was homogeneous without the application of loads, and to avoid the influence on the application of subsequent loads, the initial ground stress field of the soil was computed using the geostatic analysis step, and a self-weight load -24.5 kN/m^3 was applied to achieve stress equilibrium in the soil–rock mass (the mass of weathered granite per unit volume was set at 2500 kg . The gravitational field strength was taken as -9.8 N/kg . Therefore, the equivalent gravity per unit volume was calculated to be -24.5 kN , meaning that the applied self-weight load was -24.5 kN/m^3). The initial ground stress contour plot and vertical displacement contour plot after equilibrium were extracted are shown in Figure 4.

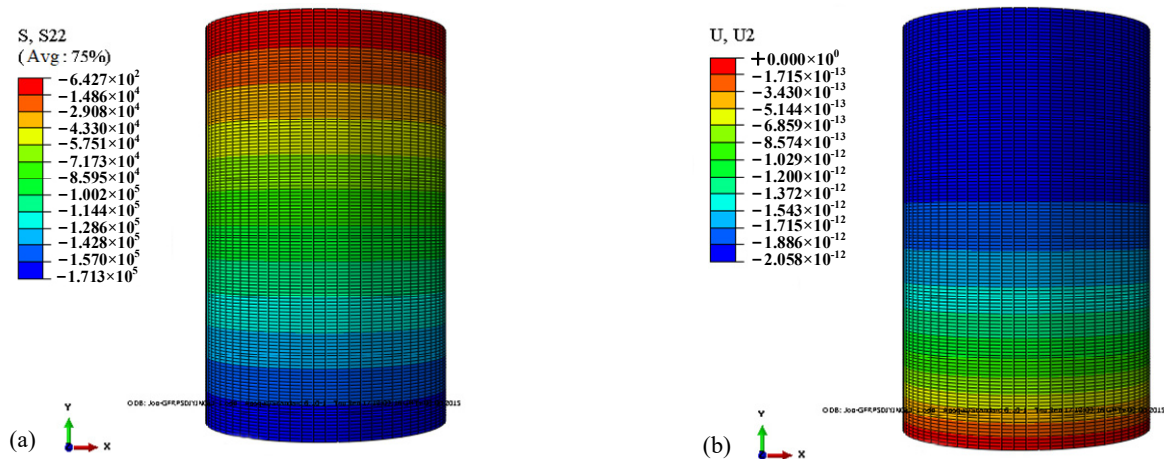


Figure 4. Initial effective stress and vertical displacement cloud: (a) vertical stress cloud after equilibrium (unit: Pa); (b) vertical displacement cloud after equilibrium (unit: m).

From Figure 4, it could be observed that the initial vertical stress is uniformly distributed at the same depth, with values increasing gradually from top to bottom. After the stress equilibrium was reached, the vertical displacement values were relatively small, with a magnitude of only 10^{-12} for the surrounding rock–soil mass. Therefore, the displacement of the surrounding rock–soil mass could be ignored for later applied loads.

4. Calculation Results and Model Validation

4.1. Load–Displacement Curve Analysis

The load–slip curves obtained from the numerical simulation and in-situ pull-out tests are shown in Figure 5.

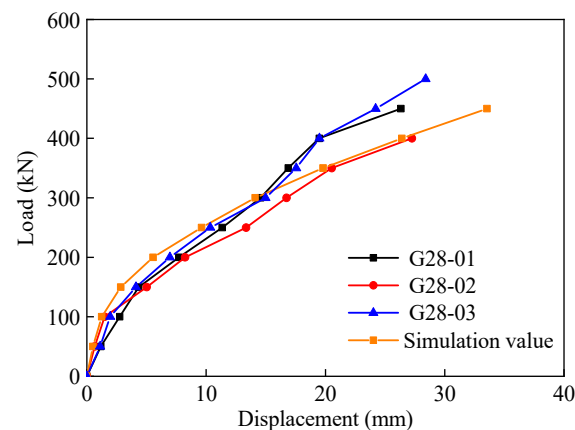


Figure 5. Load–displacement curve comparison.

From Figure 5, it could be observed that the overall trends of the simulated and test curves were similar; both showed a gradual variation type. Initially, the displacement of the test anchor was slightly larger than the simulated calculation value. However, as the load increased, the simulated result gradually exceeded the test values. The maximum failure load of the simulated anchor rod was 450 kN, with a total simulated displacement of 33.54 mm, which was greater than the test value of 28.4 mm. The errors between the experimental results and simulations were 18.10%, which fell within the acceptable range of error. The reasons for this were as follows: (1) The numerical simulation results were based on an ideal homogeneous environment, and at this stage, the load applied to the test anchor rod was relatively small, so the effect of the loading method on the confining pressure of the anchor was not significant. Therefore, under the combined effect of various factors, the displacement of the test anchor was relatively large. However, as the load continued to increase, the loading method in the in-situ test became more significant in terms of confining pressure on the anchor, resulting in an increase in frictional force between the anchor rod body and the surrounding anchor grout and thus a relatively smaller displacement. (2) The numerical simulation results represent the final slip at each load level, while the test results represent the slip within a certain time (2 h), which accumulates over time and increased continuously. However, the total displacement in the simulation is similar to the total displacement in the test, which indicated a good agreement between the numerical simulation results and the test results, thus validating the accuracy of the model.

The displacement of the anchor head for all three tested anchors and the simulated displacement increased with the increase of the pull-out load. The ultimate failure load for all rods exceeded the theoretical design value of 342 kN. When the load level was relatively low, i.e., $Q \leq 150$ kN, the load and slip values for both the test and simulated results approximately followed a linear relationship. As the load continued to increase, the relationship between load and slip gradually became nonlinear. However, compared to the test results, the simulated results exhibited a more uniform and regular variation. This phenomenon was mainly due to the complexity of the in-situ test process, which was easily influenced by various uncertain factors such as test materials, spatiotemporal effects, and groundwater conditions.

4.2. Anchor Rod Body Axial Forces Distribution

4.2.1. Distribution of Axial Forces in the Body of Anchor Rods

The variation curve of axial stress along the depth direction of the anchor was plotted, and the axial stress distribution curve along the depth direction of the anchor was calculated using a numerical simulation, as shown in Figure 6.

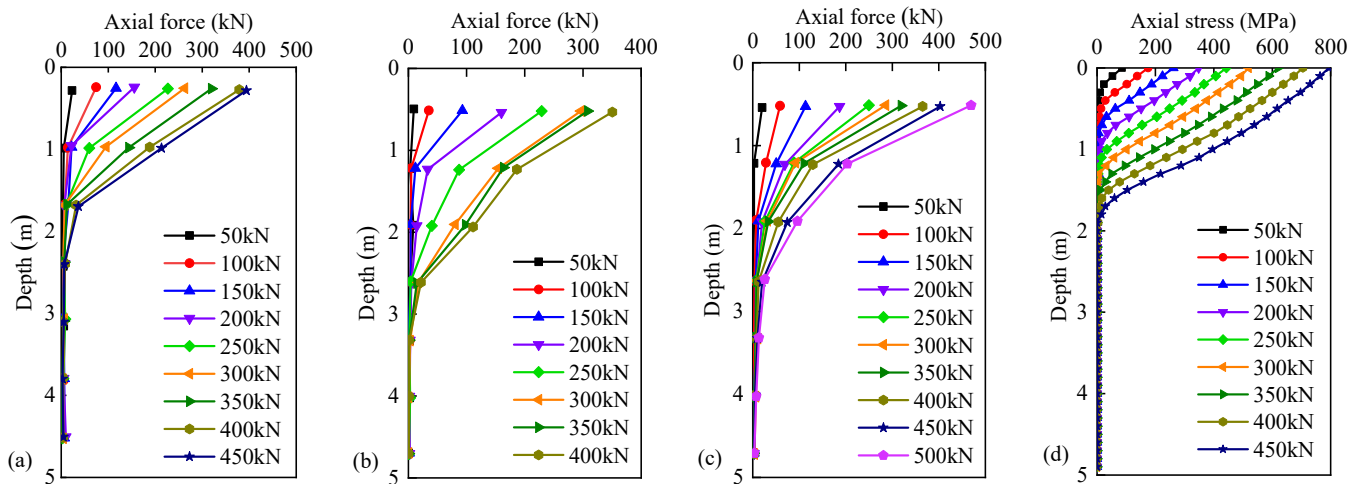


Figure 6. Variation curves of anchor shaft force along depth: (a) G28-01; (b) G28-02; (c) G28-03 Variation curve of anchor shaft force along depth. (d) Variation curve of anchor shaft stress along depth for simulated anchors.

From Figure 6a–c, it could be observed that the variation patterns of axial stress along the depth of the three tested anchors were consistent. With the increase of load, the depth of axial stress transfer gradually increased and then decreased with depth. Furthermore, as the load continued to increase, the increment of axial stress decreased with increasing depth. In other words, the maximum increment occurred at the opening of the hole, and the trend of axial stress increase became smaller as it went deeper. The distribution of axial stress along the depth direction of the GFRP anti-floating anchor was not uniform, but gradually transferred downward until the axial stress became zero. In summary, the axial stress of the GFRP anti-floating anchor gradually transferred downward along the depth of the anchor, showing a “decay” state. From Figure 6d, it could be observed that the axial stress transfer obtained from the numerical simulation is similar to the in-situ test. However, due to the idealized conditions in the numerical simulation, the curve of axial stress variation was smoother and more regular compared to the in-situ test, especially under high load levels. The simulated axial stress distribution curve showed that, at lower loads, the axial stress distribution followed a negative exponential distribution and transferred downward. As the load level increased, the range of maximum axial stress in the anchor tended to expand, reflecting the diffusion characteristic of axial stress under high load levels.

4.2.2. Critical Anchoring Length

The critical anchorage length of anchor structures referred to the maximum anchorage length of anchors in a certain rock–soil medium. Beyond this length, the bearing capacity will no longer increase significantly. Below this length, the bearing capacity had a certain safety margin [15]. For the critical anchorage length, the in-situ tests of three anti-floating anchors determined that the critical anchorage length was approximately 3.5 m deep from the ground surface. This means that the axial stress gradually transferred downward along the depth direction, and after reaching a depth of 3.5 m, the axial stress on the anchor was basically zero. This indicated that the part below 3.5 m from the opening had minimal influence on the bearing capacity of the entire anchor. Therefore, simply increasing the anchorage length cannot effectively improve the pull-out bearing capacity of the anchor.

Combined with the axial stress distribution obtained from the numerical simulation, it was found that the depth of influence of the axial stress from the numerical simulation was approximately 2.5 m. This means that the axial stress in the anchor below a 2.5 m depth was basically zero. Especially at the final load level of 450 kN, the axial stress at a depth of 3.0 m was only 0.02 kN. The reasons for this were analyzed as follows: (1) The numerical simulation idealized the conditions and parameters compared to the in-situ tests, such as not considering the bonding defects that might occur during the grouting process. (2) The numerical simulation did not account for the influence of cement grout solidification shrinkage, resulting in a better bonding performance of the cement grout in the numerical simulation.

Both the numerical simulation results and in-situ test results demonstrated the concept of a critical anchorage length, which means the depth below a certain anchorage length did not contribute to the bearing capacity of the anchor. Therefore, considering the different construction methods of anchors, different design details, and local variations in the rock–soil mass, the anchorage length should not exceed 5.0 m. The length of the anchorage section of the anchor should not be too short either. It should not only ensure the sufficient development of bonding stress between the surrounding rock–soil mass and the anchor body but also provide enough stress reserve in the anchor to ensure the overall stability of the anti-floating structure. The results of this test indicated that, under the condition of moderately weathered granite, the reasonable anchorage length of the 28 mm-diameter GFRP anti-floating anchor was between 3.5 and 5.0 m. However, compared to the numerical simulation results, at a load level of 450 kN, the load at a depth of 3.0 m was only 0.02 kN, which suggested that the simulated calculation underestimated the depth of stress transfer in the anchor.

4.3. Anchor Rod Body Shear Stress Distribution

The variation curve of shear stress along the depth direction of the GFRP anti-floating anchor was plotted based on the shear stress calculated from in-situ tests and the shear stress obtained from the numerical simulation, as shown in Figure 7.

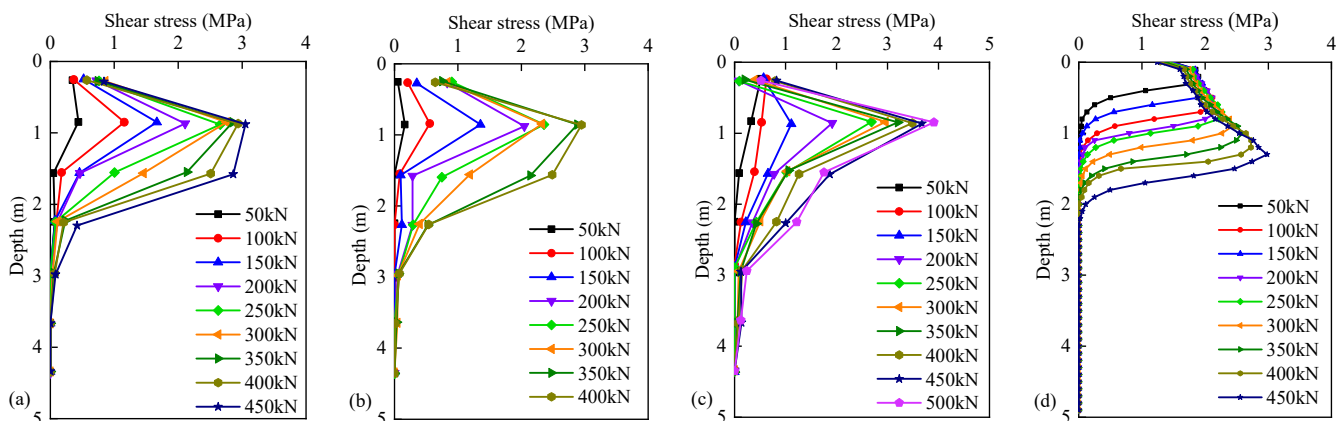


Figure 7. Variation curve of anchor shear stress along the depth direction: (a) G28-01; (b) G28-02; (c) G28-03 anchor shear stress variation curve along depth; (d) simulated anchor shear stress variation curve along depth.

From Figure 7a–c, the shear stress of the three tested anchors rapidly increased from zero at the borehole opening to a maximum value, and then decreased rapidly. It tended to zero at a certain depth. The maximum shear stress under each load level occurred at a depth of approximately 1.0 m in the anchor. Furthermore, with the increase in load, the peak shear stress gradually increased, reaching a maximum shear stress of 4.0 MPa. However, from Figure 7d, it could be observed that the depth position of the maximum shear stress under different load levels generated by the numerical simulation continuously changed and moved towards the depth direction of the anchor with increased load levels. Nevertheless,

the trend of shear stress variation obtained from both the test and numerical simulation was similar, showing an initial increase followed by a decrease. Similarly, with the increase in load level, the shear stress at the borehole opening first increased and then decreased.

Both the test and simulation results showed the presence of shear stress peak points. However, the shear stress obtained from the test represents the average shear stress, which resulted in different trends compared to the calculated shear stress. Nevertheless, through analysis, it could be concluded that the result obtained from the simulation, where the shear stress peak point gradually moved deeper with an increased load, exhibited a more regular pattern. As the load increased, the bonding interface between the anchor rod body and the anchoring mortar was gradually overcome by tensile forces, leading to the softening and decoupling of the bonding interface at shallow depths. This results in the downward shift of the shear stress peak point. According to the simulation results, the shear stress peak point under a load level of 450 kN was located at a depth of 1.2 m in the anchor rod body.

In conclusion, it was found that the shear stress transfer depth of the anchor rod body in the three tested specimens was approximately 3.5 m. After reaching a depth of 3.5 m, the shear stress on the anchor was essentially zero. The numerical simulation results showed that the shear stress transfer depth was approximately 2.5 m, meaning that the shear stress in the rod body was basically zero below a depth of 2.5 m. Based on the distribution pattern of the shear stress in the rod body, the reasonable anchorage length of the GFRP anchor under this test condition was between 3.5 and 5.0 m. This was consistent with the critical anchorage length determined by the distribution pattern of axial force in the rod body, further confirming the existence of the critical anchorage length and the reliability of the reasonable anchorage length.

5. Discussion

The GFRP anti-floating anchor anchorage system consisted of the anchor rod body, anchor mortar, and rock–soil mass, as well as the bonding interface between the anchor rod body and the anchor mortar (first interface) and the anchor mortar and the rock–soil mass (second interface). The anchorage system was shown in Figure 8. Currently, research on the anchorage system has mainly focused on the anchor rod body itself. However, relevant research results indicated that the phenomenon of rod body fracture or failure was rare due to the fact that the anti-pull-out strength of the anchor rod body was smaller than the ultimate pull-out force. In practical engineering, the failure of the anchorage system mostly occurred between the anchor rod body and the anchor mortar, as well as between the anchor mortar and the surrounding rock–soil mass. However, the change in stress and displacement of the anchor mortar and the surrounding rock–soil mass, which could not be measured by in-situ ultimate pull-out tests, remain unknown. Therefore, in order to clarify the scope of influence of the GFRP anti-floating anchor after being subjected to load and to have a more comprehensive understanding of the stress characteristics of the anchorage system, this study was based on the aforementioned numerical model to investigate the stress distribution of the anchor mortar and the stress and displacement distribution of the surrounding rock–soil mass under different load conditions.

5.1. Characterization of Stress Distribution in Anchored Mortar

Stress distribution of anchor mortar at different load levels based on simulation results: stress contour map of anchor mortar extracted at various load levels, as shown in Figure 9.

From Figure 9, it could be observed that the stress of the anchor mortar gradually increased and transmits deeper as the load increased. The anchor mortar gradually exerted its anchoring effect from top to bottom. At a load of 50 kN, the maximum stress of the anchor mortar was located at the anchor head, with a value of 2.6 MPa. As the load increased to 450 kN, the maximum stress position shifted to a depth of 1.0 to 2.0 m below the anchor head, with a maximum stress value of 3.9 MPa. Compared to the 50 kN load, the stress amplitude increased by 50% under the 450 kN load. The stress transfer efficiency between the GFRP anchor body and anchor mortar was relatively low. When the load

on the rod body increased by nine times, the maximum stress increment in the anchor mortar was only 50%. Moreover, all in-situ test anchors exhibited interface shear slip failure between the rod body and the anchor mortar, indicating that the bonding performance between the mortar anchor filler and GFRP was not excellent. However, considering that the average ultimate failure load of the three tested anchors and the maximum simulated failure load of the anchors are both 450 kN, which was higher than the theoretical design value of 342 kN, with an increment of 31.58%, it could be concluded that the mortar anchor filler could meet the engineering requirements. The reason for this phenomenon could be attributed to the gradual softening and decoupling of the bonding interface between the anchor mortar and the anchor rod body along the depth direction as the load increased. This leads to the downward shift of the stress extreme point. With the increase in the range of softening and decoupling, the stress transfer efficiency between the anchor rod body and the anchor mortar decreased. Therefore, when the load on the rod body increased by nine times, the maximum stress increased in the anchor mortar was only 50%. When the load increased to a sufficiently large value and the anchor rod body had not reached its ultimate pull-out strength, inadequate bonding occurred between the anchor rod body and the anchor mortar at the location of maximum shear stress. This led to the failure of anchor rod pull-out, also known as the first interface failure.

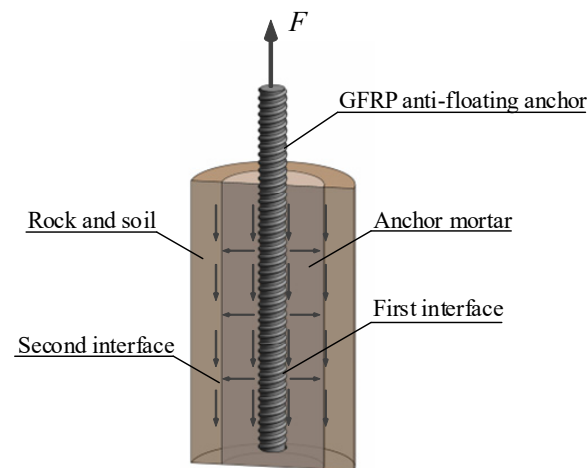


Figure 8. Physical model of GFRP anti-floating anchor anchorage system. (F is the tensile force applied at the top of the anchor rod body).

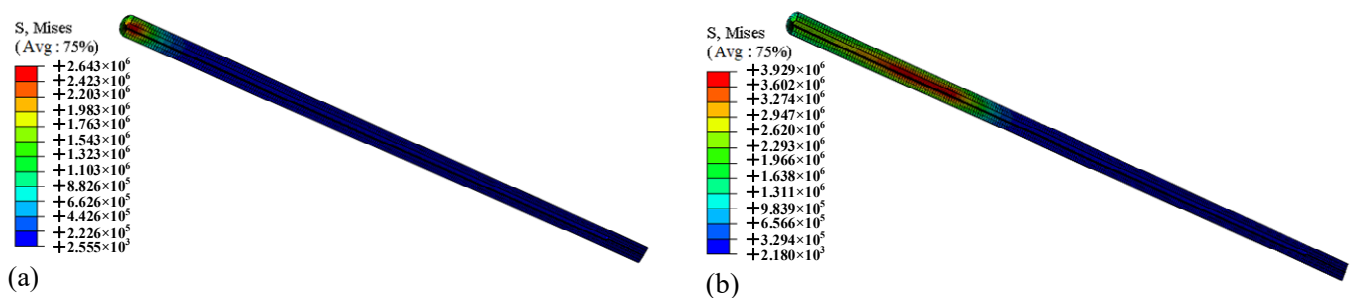


Figure 9. Stress contour map of anchor mortar: (a) stress contour map of mortar at a load of 50 kN (unit: Pa); (b) stress contour map of mortar at a load of 450 kN (unit: Pa).

5.2. Stress and Displacement Distribution Characteristics of Rock–Soil Mass

Stress and displacement contour maps of the surrounding soil at a load of 450 kN were extracted, as shown in Figure 10.

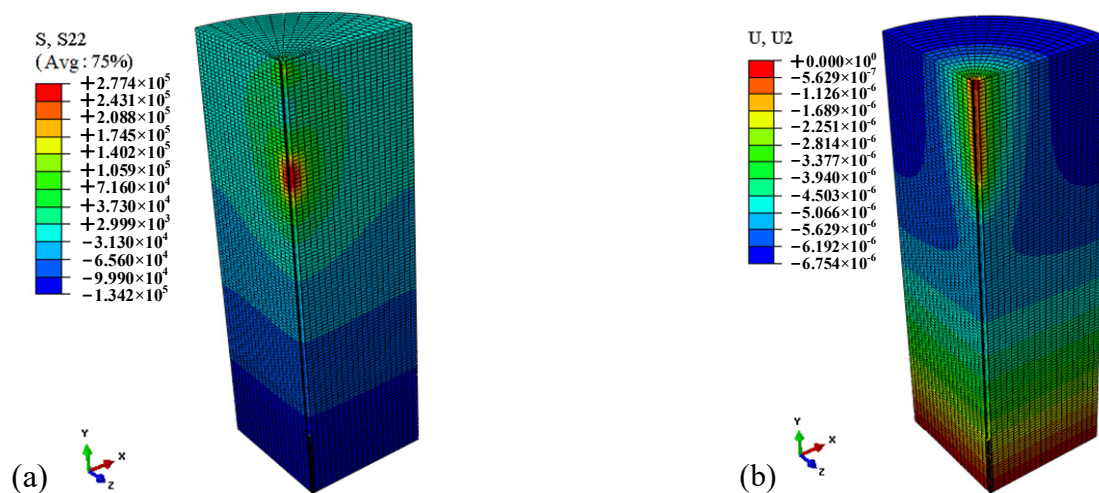


Figure 10. Vertical stress and displacement contour maps of the surrounding rock–soil mass at a load of 450 kN: (a) vertical stress contour map of the surrounding rock mass (unit: Pa); (b) vertical displacement contour map of the surrounding rock mass (unit: m).

From Figure 10a, it could be observed that when the load was 450 kN, the lateral influence range of the GFRP anti-floating anchor anchoring system on the surrounding rock–soil mass was a circular area with a radius of 1.75 m. The vertical influence depth was approximately 3.0 m, and the maximum stress position was located at a depth of 1.0 to 2.0 m from the anchor head, corresponding to the location of the maximum stress in the anchor rod body and the anchor mortar. The maximum stress value was 0.28 MPa. From Figure 10b, it could be seen that the displacement changed of the surrounding rock–soil mass due to the tension and pull-out of the anchor system, which exhibited a decreasing trend from the inside to the outside and from top to bottom. However, the impact on the anchor system was minimal, with a magnitude of approximately 10^{-6} . Therefore, compared to the slip displacement between the anchor rod body and the anchor mortar, the displacement of the surrounding rock–soil mass could be neglected and remains in a static state. As the load continued to increase, the anchor rod body had not yet reached its ultimate pull-out strength, and there was no inadequate bonding between the anchor rod body and the anchor mortar. The bond strength between the anchor mortar and the surrounding rock–soil mass reached its limit first, leading to the pull-out failure of the anchor mortar, also known as the second interface failure.

In conclusion, as the load increased, the stress in the anchor mortar gradually increased. However, the stress transfer efficiency decreased as the softening and decoupling range of the bonding interface between the anchor rod body and the anchor mortar increased, resulting in a downward shift of the stress peak. Ultimately, the insufficient bonding strength between the anchor rod body and the anchor mortar led to interface shear slip failure at the location of the maximum shear stress between them, revealing the essence of GFRP anchor failure from a mechanical perspective. On the other hand, the influence on the surrounding rock–soil mass exhibited a decreasing trend from the inside to the outside and from top to bottom. However, the impact on the anchor system was minimal. This failure required that the anchor rod body had not yet reached its ultimate pull-out strength and there was no inadequate bonding between the anchor rod body and the anchor mortar. Additionally, the bond strength between the anchor mortar and the rock–soil mass reached its limit first, leading to the pull-out failure of the anchor mortar. However, compared to steel reinforcement, GFRP reinforcement had a smaller elastic modulus, which resulted in larger slip displacement with the anchor mortar in pull-out tests. Combined with the stress distribution characteristics of the anchor mortar obtained from numerical simulations, it was confirmed that the reason for the occurrence of interface shear slip failure between the GFRP anchor rod body and the anchor mortar was due to the properties of GFRP anchors.

6. Conclusions

In this study, a finite element software (Abaqus/CAE 2021) was used to create a computer model of GFRP anti-floating anchors in weathered granite. In conjunction with in-situ ultimate pull-out tests, the bonding anchoring performance and bearing characteristics between the anchor rod body, anchor mortar, and rock–soil mass were investigated. The essence of anchor failure was revealed from the perspective of the stress and displacement distribution characteristics of the anchor mortar and rock–soil mass. Based on the results, the following conclusions were drawn:

1. Cohesive bonding elements showed a high level of agreement in simulating the interface relationships within the GFRP anti-floating anchor anchoring system. Using this element to define the interface contacts between the anchor rod body and the anchor mortar as well as between the anchor mortar and the rock–soil mass could effectively reflect the actual bonding behavior of the anchor.
2. The axial stress in the GFRP anti-floating anchor rod body attenuates along the depth direction, and there existed a critical value for the anchor length. Under the same conditions, a reasonable anchoring length should be between 3.5 to 5.0 m. The simulated shear stress peak position of the anti-floating anchor rod body exhibited a phenomenon of moving towards the deeper region.
3. The stress in the anchor mortar gradually increased and transferred to the deeper region as the load increased. The anchor mortar gradually fulfilled its anchoring function from top to bottom. As the load increased from 50 kN to 450 kN, the maximum stress in the anchor mortar showed an amplification of 50%. The bonding performance between the cement mortar filler and GFRP anchor was not excellent, but it could meet the engineering requirements.
4. Under a load of 450 kN, the lateral influenced range of the anchoring system on the rock–soil mass was a circular area with a radius of 1.75 m, and the vertical influence depth was approximately 3.0 m. The displacement change on the surrounding rock and soil mass gradually decreased from the inside to the outside and from the top to the bottom, and the overall displacement of the rock and soil mass was relatively small.

Author Contributions: Conceptualization, H.L.; Methodology, Z.W.; Software, N.Y.; Validation, J.H.; Formal analysis, C.M.; Investigation, S.J.; Data curation, L.Z.; Writing—original draft, Z.H.; Writing—review & editing, N.Y., X.B., G.S. and M.Z.; Funding acquisition, M.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by the National Natural Science Foundation of China (Grant no. 51708316), the Key Program of Natural Science Foundation of Shandong Province (Grant No. ZR2020KE009), Key Laboratory of Geological Safety of Coastal Urban Underground Space, Ministry of Natural Resources, Open Fund Key Project (BHKF2021Z09), Taishan Scholars (NO.tsqn202306234) supported by Shandong Province, and the China Postdoctoral Science Foundation Funding (2018M632641).

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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